Tunnelling and Underground Space Technology 171 (2026) 107457

Contents lists available at ScienceDirect

Tunnelling and Underground Space Technology incorporating Trenchless Technology Research

journal homepage: www.elsevier.com/locate/tust

Digital twin-based framework for multi-objective optimization of shield tunneling parameters☆

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ARTICLE INFO

Keywords:
Shield tunneling
Digital twin
Parameter optimization
Multi-objective optimization

ABSTRACT

Optimizing shield tunneling parameters is critical for ensuring both safety and efficiency in tunnel construction. However, existing optimization approaches often underutilize operational data and are prone to entrapment in local optima. This study proposes a digital twin (DT) -based framework for multi-objective optimization (MOO) of shield tunneling parameters. The framework integrates data acquisition, preprocessing, modeling, and optimization within a layered architecture. Key parameters are identified using the shapley additive explanations (SHAP) method, while a hybrid optimization algorithm—artificial bee colony (ABC)–non-dominated sorting genetic algorithm III (NSGA-III)—combines the strengths of both algorithms and is applied across three operational scenarios. Optimized parameters are iteratively fed back into the DT to guide parameter adjustment. The framework is validated using data from the Shanghai Airport Link Line project. Full-parameter optimization yields the best performance, achieving an overall optimization rate of 32.02%, with particularly notable improvements in controlling the vertical deviation of shield head. Comparative analyses show that the proposed framework surpasses benchmark methods in convergence speed and solution quality, reducing shield attitude deviation by 2.21%–17.13%. These results underscore the framework's potential as an effective decision-support tool for shield tunneling operations.

1. Introduction

With the rapid expansion of urban underground infrastructure, the demand for tunnel construction has increased substantially (Guo et al., 2025). Among the available tunneling methods, shield tunneling has been widely adopted due to its advantages in construction efficiency, operational safety, and minimal disturbance to surrounding environments (Lei et al., 2024). Nevertheless, shield tunneling control remains inherently complex, and inappropriate tunneling parameter settings can induce deviations in shield attitude, leading to trajectory misalignment, segment dislocation, or even structural damage (Li et al., 2024). Therefore, optimizing shield tunneling parameters is essential to maintain alignment with the designed tunnel axis and ensure construction safety (Shen et al., 2022).

Optimization of shield tunneling parameters involves multiple interdependent variables—such as thrust force, cutterhead torque, and advance speed—that exhibit nonlinear relationships (Fu et al., 2026). As such, the problem is typically formulated as a multi-objective optimization (MOO) task (Chen et al., 2024). Existing approaches to addressing this problem include statistical analysis, numerical simulation, and intelligent algorithms (Chang et al., 2024). Statistical analysis methods can reveal empirical relationships among variables (Sugimoto and Sramoon, 2002), but often fail to capture nonlinear interactions under heterogeneous geological conditions (Huang et al., 2022). Fu et al. (2025) introduced automated stratum identification techniques to address challenges arising from geological uncertainties. Numerical simulations, such as those based on finite element methods, provide valuable insights into mechanical behavior and deformation control (Yin et al., 2018), yet their applicability is limited by computational cost (Huang et al., 2024). More recently, intelligent optimization algorithms

☆ This article is part of a special issue entitled: 'Advances in Shield Tunneling' published in Tunnelling and Underground Space Technology incorporating Trenchless Technology Research.

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https://doi.org/10.1016/j.tust.2026.107457
Received 18 August 2025; Received in revised form 22 November 2025; Accepted 6 January 2026
Available online 13 January 2026
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have demonstrated strong optimization capabilities.

Several MOO algorithms such as non-dominated sorting genetic algorithm II (NSGA-II) (Deb et al., 2002), non-dominated sorting genetic algorithm III (NSGA-III) (Deb and Jain, 2014), multi-objective evolutionary algorithm based on decomposition (MOEA/D) (Fan et al., 2022; Wang et al., 2021), and multi-objective marine predator algorithm (MOMPA) (Zhong et al., 2021), have been developed and applied across engineering domains. These algorithms are well-suited for problems characterized by nonlinear relationships, high dimensionality, and conflicting objectives. However, when dealing with six or more objectives, MOEA/D struggles to maintain a uniform distribution of weight vectors, whereas MOMPA tends to converge prematurely to local optima due to limited mechanisms for preserving population diversity (Lai et al., 2025). By contrast, NSGA-III employs a reference-point-based selection strategy that effectively preserves solution diversity in high-dimensional objective spaces, rendering it suitable for complex MOO problems. In the context of shield tunneling, NSGA-III has been applied to optimize critical tunneling parameters. For instance, Liu et al. (2021) utilized it to determine optimal cutterhead rotation speeds. Nevertheless, NSGA-III exhibits slow convergence, which may hinder its ability to approximate the true Pareto front (Cui et al., 2019). Thus, a research gap remains in achieving both convergence performance and solution quality in high-dimensional optimization for tunneling parameters.

To address these limitations, *meta*-heuristic algorithms such as particle swarm optimization (Tharwat and Schenck, 2021), ant colony optimization (Priyadarshi and Kumar, 2025), and artificial bee colony (ABC) optimization (Ibrahim et al., 2025) have been used to navigate complex search spaces and avoid entrapment in local optima (Hussain et al., 2020). Among them, the ABC algorithm is particularly notable for its strong global search capability, inspired by the collective foraging behavior of honeybees (Lin et al., 2025). Accordingly, integrating the ABC algorithm with NSGA-III offers the potential to improve the diversity and quality of initial candidate solutions, thereby enhancing the overall optimization performance. However, current research on shield tunneling parameter optimization remains predominantly algorithm-centric, with limited attention given to integrating optimization models with the underlying physical processes of tunneling (Li et al., 2019).

In response, the emergence of digital twin (DT) technology has attracted growing attention as a promising pathway toward bridging this gap (Kritzinger et al., 2018). First conceptualized by Grieves (2005), DT technology establishes high-fidelity virtual representations of physical assets, enabling interactions among simulation, optimization, and feedback. This capability creates new opportunities for intelligent control in tunnel construction (Barata and Kayser, 2024). For example, Ye et al. (2023) developed a DT-based safety management system for simulating tunneling processes and monitoring risks; and Feng et al. (2024) applied DT to simulate ground deformation. Despite these advances, most DT applications focused on simulation and monitoring, with limited functionality for high-level optimization decisions (Zheng et al., 2024).

This study proposes a dynamic optimization framework for shield tunnelling parameters by integrating DT with an improved NSGA-III algorithm. Leveraging the continuous synchronization capabilities of DT, data streams are used to update constraint conditions and objective functions, which overcome the shortcomings of conventional static decision-making approaches. The main contributions of this study are as follows: (1) A dynamic optimization framework that integrates DT technology with an enhanced evolutionary algorithm is developed, expanding the application scope of DT in tunnel construction and advancing its functional evolution from monitoring and simulation to intelligent decision-making; (2) A human–machine collaborative closed-loop architecture is introduced, which enhances the framework's adaptability under complex tunneling conditions by integrating algorithmic optimization with expert operational knowledge.

In the remainder of this paper, Section 2 presents the DT framework and the MOO approach for shield tunneling parameters. Section 3 evaluates the performance of the proposed framework through a real-world case study. Section 4 discusses the case study results. Section 5 concludes this study.

2. Framework for tunneling parameter optimization

2.1. Overview of the framework

As illustrated in Fig. 1, the proposed DT-based tunneling parameter optimization framework comprises four interrelated layers: the physical layer, responsible for monitoring of the tunneling process; the virtual layer, dedicated to constructing a high-fidelity DT model; the data interaction layer, which ensures seamless synchronization between the physical and virtual layers; and the application service layer, which executes dynamic optimizations and generates operational feedback. Grounded in continuously synchronized data, the integration of these layers forms a closed-loop architecture enabling parameter perception, intelligent optimization, and dynamic feedback. This integrated design enables a paradigm shift from passive monitoring to data-driven, human–machine collaborative shield tunneling operations.

Effective optimization of shield tunneling parameters necessitates a robust mapping mechanism that connects physical measurements with virtual modeling. The physical layer comprises the shield machine and an array of data acquisition devices, which together provide the primary data source for parameter optimization. After data preprocessing and analysis, tunneling parameters and geological information are extracted as input variables for the MOO algorithm. The multi-dimensional nature of these variables directly affects the interdependencies among the objective functions that must be balanced during optimization. Table 1 summarizes the types of sensors used to quantify tunneling performance.

The virtual layer establishes a digital environment for parameter optimization through the coordinated integration of three models. The geometric model provides spatial constraints for shield machine. The behavior model uses the shapley additive explanations(SHAP) method (Qiao et al., 2024) to construct a dynamic response mechanism between tunneling parameters and optimization objectives. The rule model defines permissible parameter ranges and operational constraints, thereby establishing a feasible decision space for the MOO problem. Interaction among these models allows the optimization algorithm to periodically evaluate the impact of different parameter combinations on conflicting objectives within the virtual environment.

The application service layer integrates monitoring data from the physical layer with virtual modeling to execute tunneling parameter optimization. By using an advanced MOO algorithm, this layer identifies optimal parameter combinations and established recommended control ranges aimed at minimizing shield attitude deviations. Once the optimization results are obtained, the geometric model within the virtual layer maps the optimized parameters to corresponding components or locations of the shield machine. These validated adjustments are subsequently transmitted back to the physical layer, ensuring adaptive control of shield advancement.

The data interaction layer serves as the information hub of the framework, enabling seamless communication and integration across all layers. It manages data flows among the physical layer, virtual models, application service layer, and embedded knowledge bases, including industry standards, optimization algorithms, and historical parameter libraries. Through multi-source data integration and transmission, this layer establishes a comprehensive data integration framework to provide data support and feedback for parameter optimization, and enhances the level of intelligence in shield tunneling operations.

2.2. Data preprocessing

2.2.1. Downtime data removal

During shield tunneling, monitoring data collected over extended

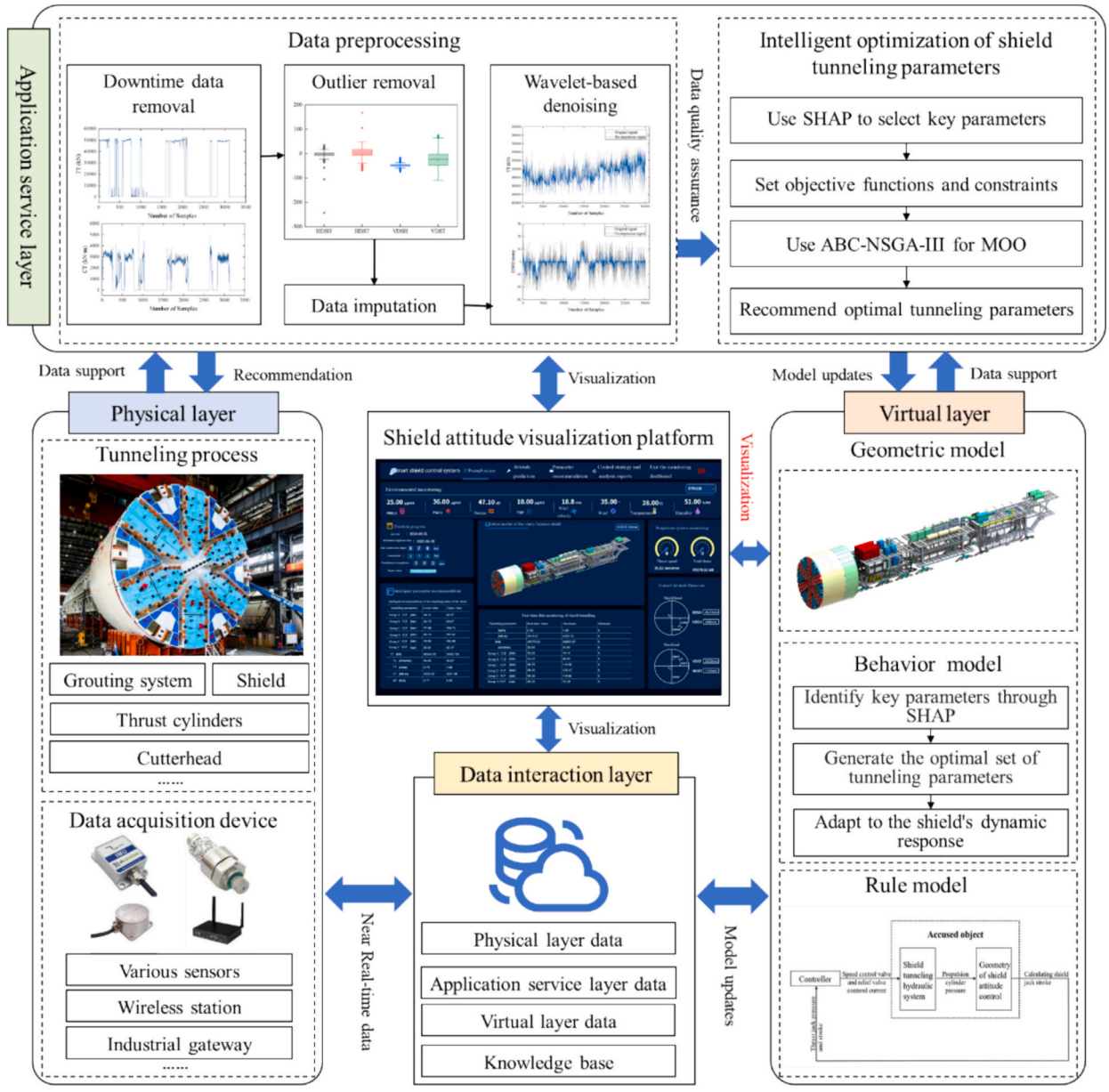


Fig. 1. Framework for shield tunneling parameter optimization.

Table 1
Common sensor types, functions, and collected data.

Type	Function	Collected data
Pressure sensor	Measure pressure at the cutterhead and key locations	Burial depth; Internal friction angle; Soil cohesion; Thrust cylinder pressure; Total thrust; Cutterhead torque; Grouting pressure; Inlet slurry density; Outlet slurry density; Air cushion pressure; Slurry pressure
Displacement sensor	Measure stroke length of thrust cylinders and other conditions.	Cutterhead penetration rate; Horizontal deviation of shield head; Vertical deviation of shield head; Horizontal deviation of shield tail; Vertical deviation of shield tail
Speed sensor	Measure advance speed, cutterhead rotation rate and other conditions.	Thrust speed; Cutterhead speed; Inlet slurry flow rate; Outlet slurry flow rate
Tilt sensor	Measure roll and pitch of the shield machine	Roll angels; Pitch angels

shutdown periods are of limited value for parameter optimization and should be excluded from analysis. Accurate identification of such shutdown conditions depends on five key operational parameters, including advancement speed (V), cutterhead rotational speed (N), cutterhead torque (T), penetration per revolution (S), and total thrust (F). These variables collectively characterize the operational status of the shield machine. To distinguish non-tunneling or maintenance conditions, a binary discriminant function can be established to determine whether all relevant parameters simultaneously approach or equal zero. The discriminant logic is formalized in Eq. (1):

$$D = V \bullet N \bullet T \bullet S \bullet F$$

$$\text{Status} = \begin{cases} Normaltunnelingstate, D \neq 0 \\ Abnormaltunnelingstate, D = 0 \end{cases} \quad (1)$$

2.2.2. Outlier removal

To ensure dataset reliability, outliers must be detected and removed. In this study, the boxplot method is employed for outlier identification

(Kim and Noh, 2025). This statistical approach defines outlier boundaries based on quartiles and the interquartile range (IQR). The procedure is as follows:

Step 1 Quartile and IQR calculation. For each input parameter, the first quartile Q_1, third quartile Q_3, and interquartile range IQR are calculated, as illustrated in Eq. (2).

$$IQR = Q_3 - Q_1 \tag{2}$$

Step 2 Outlier range determination. The upper and lower limits for acceptable values are defined using Eq. (3). Values outside this range are considered outliers.

$$\begin{cases} lowerlimits = \mathrm{Q}_1 - 1.5 \times IQR \\ upperlimits = \mathrm{Q}_3 + 1.5 \times IQR \end{cases} \tag{3}$$

Step 3 Outlier removal. Data points beyond the defined range are removed, and only values within the range are retained for subsequent analysis.

2.2.3. Data imputation

Removing outliers can introduce missing values that must be handled to preserve dataset integrity. Simply discarding incomplete records may compromise representativeness. To address this issue, random forest (RF) imputation is applied (Luo et al., 2025). As a non-parametric, supervised learning-based approach, RF imputation leverages inter-variable correlations to estimate missing values, making it effective for high-dimensional engineering datasets characterized by nonlinear dependencies.

2.2.4. Wavelet-based denoising

Operational data collected from shield machines are often affected by environmental noise and sensor inaccuracies. To enhance signal fidelity, denoising is performed after missing-value imputation. In this study, the discrete wavelet transform (DWT) is adopted, with the Daubechies wavelet family selected as the basis function due to its strong time–frequency localization characteristics and suitability for processing non-stationary signals (Barjasteh et al., 2024).

2.3. Optimization of tunneling parameters

To address the high-dimensionality and non-linearity inherent in tunneling parameter optimization, the ABC-NSGA-III algorithm was developed within the application service layer of the proposed framework. This hybrid algorithm incorporates the global search capability of the ABC algorithm into the initialization and evolutionary processes of NSGA-III, thereby mitigating the tendency of standard NSGA-III to converge prematurely to local optima. By maintaining population diversity, ABC-NSGA-III enables a more efficient and stable search of the Pareto front. Once the Pareto-optimal solutions are obtained, the ideal point method (Xu et al., 2018) is applied to determine the most suitable combination of tunneling parameters. Finally, the optimized parameters are transmitted to the physical layer, establishing a closed-loop architecture of perception, optimization, and feedback.

2.3.1. Objective function and constraints

The optimization process is formulated to minimize six attitude-related objective functions $f_i(x)$, including horizontal deviation of shield head (HDSH), vertical deviation of shield head (VDSH), horizontal deviation of shield tail (HDST), vertical deviation of shield tail (VDST), roll angels (ROLL) and pitch angels (PITCH). The operational constraints for these objectives are defined by the national standard *Acceptance Specification for Shield Tunnel Construction* (GB50446-2017). The standard stipulates maximum allowable limits of ± 50 mm for translational deviations (HDSH, VDSH, HDST, VDST) and ± 30 mm/m for angular deviations (ROLL, PITCH). To enhance operational safety, a more stringent on-site warning threshold is set at 80% of these limits. Once the deviation of any parameter exceeds the threshold, tunneling parameters must be promptly adjusted to restore control (Wu et al., 2024). Subject to these limits and other construction-related constraints, the general mathematical formulation of this MOO problem is given by Eq. (4):

$$\begin{cases} minf_i(x) = \left(f_1(x), f_2(x), f_3(x), f_4(x), f_5(x), f_6(x)\right) \\ \quad s.t.g(x) = \left(g_1(x), g_2(x), \cdots\right) < g_{max} \end{cases} \tag{4}$$

where $g(x)$ is the constraint function of tunneling parameters;g_{max} is the corresponding upper bound.

2.3.2. Hyparameter setting

Before conducting MOO, it is necessary to reasonably configure the algorithm to ensure both convergence efficiency and global search capability. The parameters considered include the number of objective functions in NSGA-III, population size, crossover probability, mutation probability, maximum number of iterations, the number of ABC iterations, and the limit on honey collection attempts for abandoning a nectar source. Common approaches to hyperparameter optimization include random search, grid search, and Bayesian optimization (Zhao et al., 2024). Among these, grid search can systematically explore the parameter space through exhaustive traversal. Therefore, the grid search method is employed to configure the hyperparameters and determine the optimal parameter ranges.

For the optimization of the six objective functions considered in this study, parameter settings were determined following the recommendations of Deb and Jain. (2014). The population size was varied within the range of 200–500, the number of iterations between 100 and 500, the crossover probability from 0.3 to 0.7 in increments of 0.1, and the mutation probability from 0.001 to 0.02 in increments of 0.001. For each parameter combination, multiple independent runs were performed, and the configuration that achieved the best average performance across these trials was selected as the optimal parameter setting.

2.3.3. ABC-NSGA-III optimization model

An initial population of tunneling parameters is generated using the ABC algorithm, which consists of employed bees, onlooker bees, and scout bees. Employed bees explore local neighborhoods and refine current solutions using Eq. (5). Onlooker bees probabilistically select solutions based on shared information provided by employed bees and exploit promising regions as per Eq. (6). Scout bees explore the search space randomly to maintain population diversity.

$$x_{ij}^{(t+1)} = x_{ij}^{(t)} + \varphi_{ij}\left(x_{ij}^{(t)} - x_{best}^{(t)}\right) \tag{5}$$

where $x_{ij}^{(t+1)}$ represents the j-th parameter value of the newly generated candidate solution; $x_{ij}^{(t)}$ represents the j-th parameter value of the i-th solution corresponding to the current employed bee; $x_{best}^{(t)}$ represents the j-th optimal parameter value of the current employed bee.; φ_{ij} is a uniformly distributed random number.

$$p(x_{ij}) = \frac{fitness(x_{ij})}{\sum_{j=1}^{N} fitness(x_{bestj})} \tag{6}$$

where $p(x_{ij})$ is the probability that x_{ij} of the i-th solution in the j-th dimension is selected by the onlooker bee; $fitness(x_{ij})$ and $fitness(x_{bestj})$ denote the fitness of the i-th solution in the j-th dimension; N represents the parameter group size.

Adaptive normalization is applied to mitigate scale differences among objectives. Reference points are computed using Eq. (7).

$$H = \left(\frac{M+p-1}{p}\right) \tag{7}$$

where M is the number of objectives, and p is the number of the

dimension partition per objective.

Optimization objectives are normalized to mitigate dimensional inconsistencies. The minimum value of objective function Z_i^{min} is first determined. The objective values $f_i(x)$ are then transformed by subtracting the ideal point, as shown in Eq. (8). These transformations generate an M − dimensional hyperplane whose intercepts a_i on each axis are used to calculate the normalized objective functions $f_i^n(x)$, as shown in Eq. (9).

$$f_i'(X) = f_i(X) - Z_i^{min} \tag{8}$$

$$f_i^n(X) = \frac{f_i'(X)}{a_i - Z_i^{\min}} = \frac{f_i(X) - Z_i^{\min}}{a_i - Z_i^{\min}}, i = 1, 2, \cdots M \tag{9}$$

After calculating the reference points, each solution is associated with the nearest reference line extended from the origin to each reference point. Individuals are selected based on the sparsity of associations to maintain diversity. This process iterates until convergence, resulting in a well-distributed set of Pareto-optimal solutions. The specific process of this algorithm is shown in Fig. 2.

2.3.4. Ideal point method

To identify the most desirable solution from the Pareto set, the ideal point method is employed. The ideal point vector Z_i is defined as:

$$Z_{\mathrm{i}} = \left(f_1^{\min}, f_2^{\min}, \cdots f_M^{\min}\right) \tag{10}$$

To measure how close Pareto solutions are to the ideal solution, this paper uses Euclidean distance D_i, as defined in Eq. (11). The solution with the smallest distance to the ideal point is chosen as the optimal tunneling parameter scheme D_{opt}, as defined in Eq. (12).

$$D_i = \sqrt{\sum_{j=1}^{M} \left(f_{ij} - f_j^{\min}\right)^2} \tag{11}$$

$$D_{opt} = min(D_i) \tag{12}$$

where f_{ij} represents the value of solution i on objective function j.

To assess the overall quality of the solution set, the normalized hypervolume (HV) is adopted as the primary performance metric (Shang et al., 2021). HV is defined as:

$$HV(P) = VOL\left(\underset{x \in P}{\cup} [f_1(x), H_1] \times \cdots \times [f_M(x), H_M]\right) \tag{13}$$

where *VOL(.)* represents the Lebesgue measure; P represents the solution set to be measured; H_M denotes the reference point.

3. Case study

3.1. Project background

The DT-based tunneling parameter optimization framework was evaluated using a real-world case study from the western section of the Shanghai Airport Link Line project. A large slurry pressure balance shield machine was employed for tunnel construction, with lining segments of 13.6 m outer diameter and 12.5 m inner diameter. As shown in Fig. 3, a continuous stretch of 100 rings (Rings 2275–2375)—located between Air Shaft No. 2 and Air Shaft No. 3—was selected for detailed analysis. This subsection features a tight horizontal curve radius of 520 m and lies in close proximity to existing metro lines. Additionally, the geological profile in this area increases the complexity of tunnel construction, which traverses interbedded strata composed of highly permeable layers of clay, silty clay, clayey silt, and silty sand. The heterogeneous distribution of these strata creates a challenging excavation environment, making the selected subsection an ideal testbed for evaluating the performance of the proposed framework.

A set of sensory devices, as detailed in Table 1, continuously recorded key operational parameters at 10-second intervals along the tunnel alignment, enabling high-resolution monitoring of tunneling operations. To mitigate multicollinearity arising from nonlinear coupling between cutterhead penetration rate and parameters such as total thrust and cutterhead torque (Lin et al., 2025), the penetration rate was excluded during model construction to enhance generalization capability. The ranges of the final input and output parameters are presented in Table 2.

3.2. Data preprocessing

Following the data preprocessing pipeline described in Section 2.2, a visualization analysis of five shield tunneling state indicators was conducted, with the first 3,500 samples displayed for clarity (see Fig. 4).

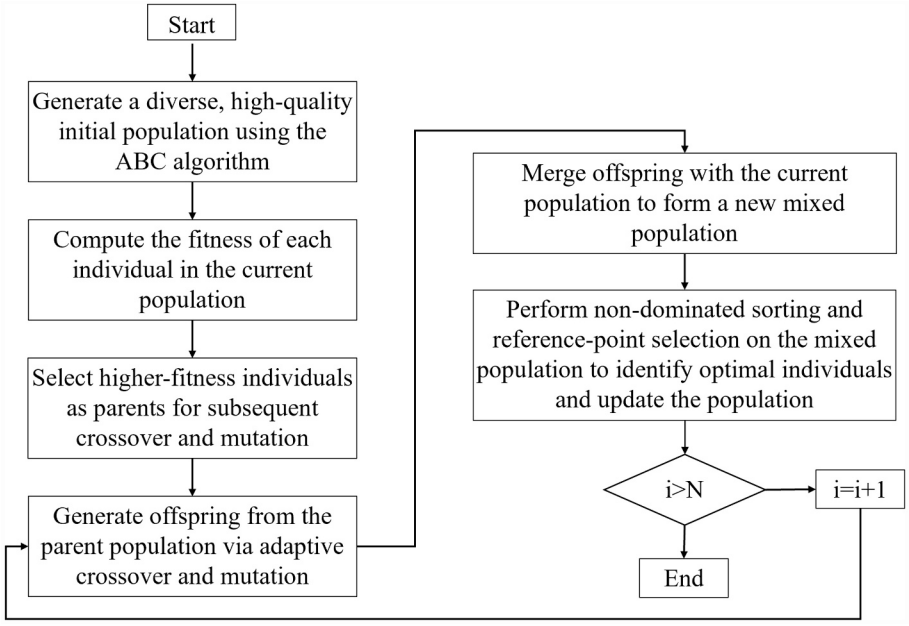


Fig. 2. Flowchart of ABC-NSGA-III.

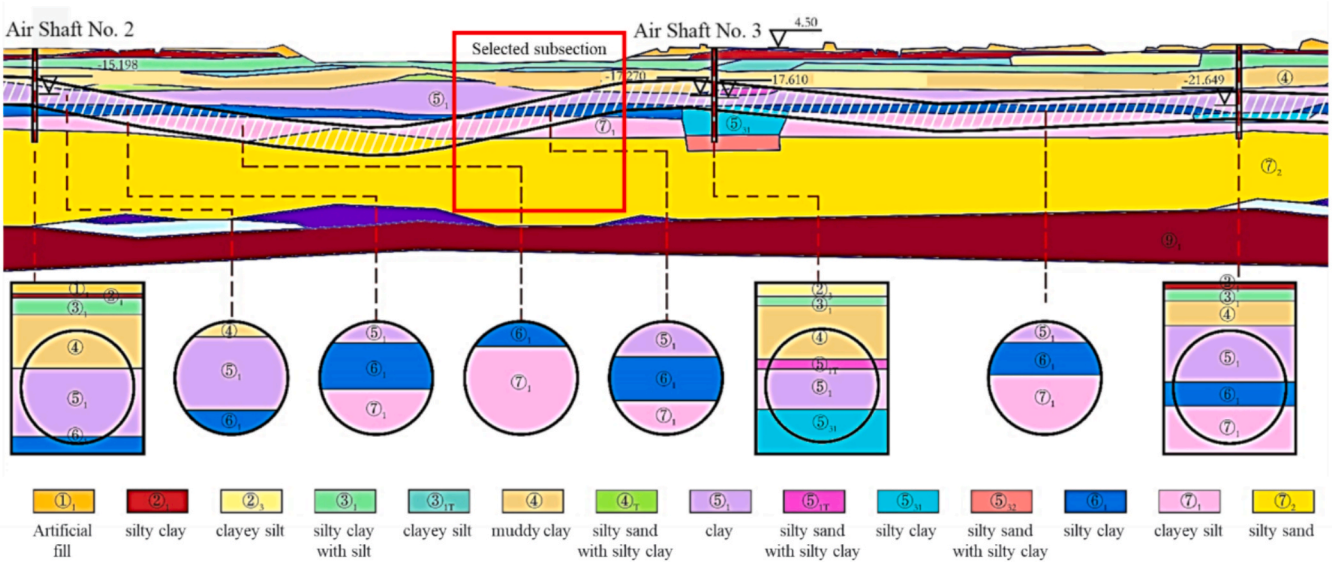


Fig. 3. Geological profile of the selected subsection.

Table 2
Range of input and output parameter values.

Parameter type	Parameter name	Max.	Min.	Ave.	Unit
Input parameter	Total thrust	54661.67	44061.14	49279.55	kN
	Cutterhead speed	1.08	0.75	0.92	r/min
	Cutterhead torque	4252.73	1482.49	2910.27	kN ·m
	Thrust speed	51.84	17.12	35.81	mm/min
	Thrust cylinder pressure-A	70.14	12.48	39.25	bar
	Thrust cylinder pressure-B	68.05	10.91	34.77	bar
	Thrust cylinder pressure-C	115.56	75.04	98.79	bar
	Thrust cylinder pressure-D	122.47	77.33	98.79	bar
	Thrust cylinder pressure-E	118.66	77.51	98.28	bar
	Thrust cylinder pressure-F	74.39	9.85	40.25	bar
	Inlet slurry density	1.34	1.30	1.31	g/cm^3
	Outlet slurry density	1.38	1.23	1.34	g/cm^3
	Inlet slurry flow rate	1870	1349	1668	m^3/h
	Outlet slurry flow rate	2169	1514	1867	m^3/h
	Grouting pressure	7.12	2.37	4.43	bar
	Air cushion pressure	3.12	2.97	3.03	bar
	Slurry pressure	3.33	2.78	3.03	bar
	Internal friction angle	9.00	21.00	16.76	°
	Soil cohesion	25.00	33.00	29.83	kPa
	Burial depth	−15.19	−34.17	−25.38	m
Output parameter	HDSH	13.52	−20.86	−1.63	mm
	HDST	45.95	−38.25	4.06	mm
	VDSH	−59.39	−34.99	−46.77	mm
	VDST	60.56	−106.30	−24.32	mm
	ROLL	1.16	−2.34	−0.46	mm/m
	PITCH	−10.12	−21.47	−15.80	mm/m

After the removal of downtime data, a total of 30,000 valid samples were retained for subsequent analysis.

Then, outliers in key parameters were removed. As illustrated in Fig. 5, the bottom and top whiskers represent the minimum and maximum values after outlier removal, the box corresponds to the IQR capturing the middle 50% of the data, and the line within the box indicates the median. Black dots outside the whiskers denote outliers, which were excluded to ensure dataset reliability. Subsequentially, the missing values introduced by this process were subsequently imputed using the RF method.

Finally, DWT-based denoising was applied to enhance data fidelity. As shown in Fig. 6, comparison before and after denoising demonstrates that the DWT can effectively suppress noise while preserving essential signal characteristics. Through this data preprocessing pipeline, a complete, structured dataset is generated.

3.3. Data transmission and visualization

The dataset from data preprocessing enables DT with perception and optimization capabilities. Within the data interaction layer of the framework, data dynamically collected from the physical layer are managed using a time-series database. Simultaneously, the virtual layer employs distributed storage systems to host geometric, behavior, and rule models—each characterized by large data volumes, complex structures, and high-frequency access requirements.

Geometric models are developed to represent the shape, dimensions, and spatial configuration of physical elements. As shown in Fig. 7(a), a high-fidelity geometric model of the slurry pressure balance shield machine was constructed using SolidWorks. In current practice, DT are predominantly developed by integrating Building Information Modeling (BIM) with sensor data, serving to provide updated representations of physical elements. The proposed framework retains these monitoring capabilities while incorporating an advanced optimization algorithm, thereby establishing a closed-loop architecture that integrates perception, optimization, and feedback. The rule model encodes constraints and control logic to ensure that the shield machine can adjust its posture under different geological conditions, while the behavior model—driven by the ABC-NSGA-III algorithm—enables adaptive control by dynamically tuning key operational parameters. Consequently, the proposed framework enables a transition from passive monitoring to proactive decision support.

All data streams are integrated within the data interaction layer and

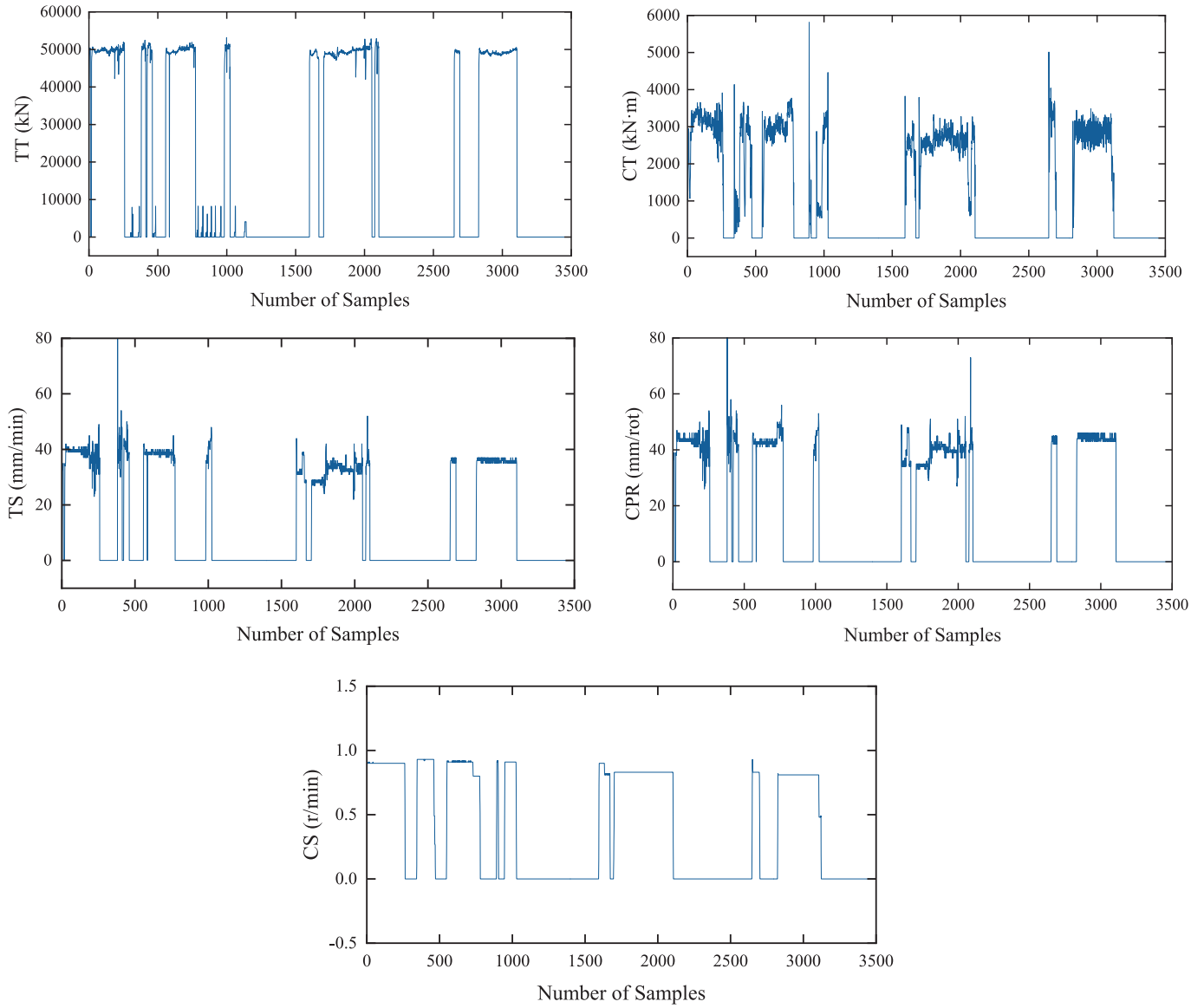


Fig. 4. Variation curves of shield tunneling state indicators.

visualized through an interactive platform, as illustrated in Fig. 7(b). This platform displays the status and spatial locations of key parameters within the shield machine, offering intuitive insights for on-site operational guidance.

3.4. Multi-objective optimization

Model training and algorithm implementation were conducted in a high-performance computing environment with the following configuration: Windows 11 (64-bit), Intel (R) Core (TM) i9-13900 K CPU @ 3.00 GHz, 64 GB RAM, using Python 3.6.13 as the programming language.

3.4.1. Feature importance analysis based on SHAP

The SHAP method was employed to assess the contribution of 20 tunneling parameters to shield attitude outcomes. This method quantifies the marginal impact of each parameter on shield attitude, as illustrated in Fig. 8. The parameters were ranked based on their average SHAP values.

The pressures from the six sets of thrust cylinder pressure exert the most significant influence on shield attitude. As primary actuators, these pressures directly determine the thrust force and operational stability of the shield machine. Their precise control is crucial for maintaining and adjusting shield machine's attitude. Parameters such as cutterhead speed, total thrust, thrust speed, cutterhead torque, air cushion pressure, inlet slurry density, outlet slurry density, inlet slurry flow rate, outlet slurry flow rate, and grouting pressure also play a vital role. Cutterhead speed and cutterhead torque are directly associated with cutting performance and equipment loading, while total thrust and thrust speed influence the advance trajectory. Parameters like air cushion pressure, inlet slurry density, outlet slurry density, inlet slurry flow rate, outlet slurry flow rate, and grouting pressure affect shield attitude indirectly through their impact on soil disturbance, cutting resistance, and slurry rheology.

The importance of different parameter categories follows the order: structural > geometric > geological. Owing to the limited variation of geometric and geological parameters in the dataset, factors such as slurry pressure, burial depth, soil cohesion, and internal friction angle exhibit relatively minor influence, indicating their limited effect on attitude control. Considering that geometric and geological parameters are typically predetermined during the design phase and remain fixed during operation, the study ultimately identified 16 key adjustable

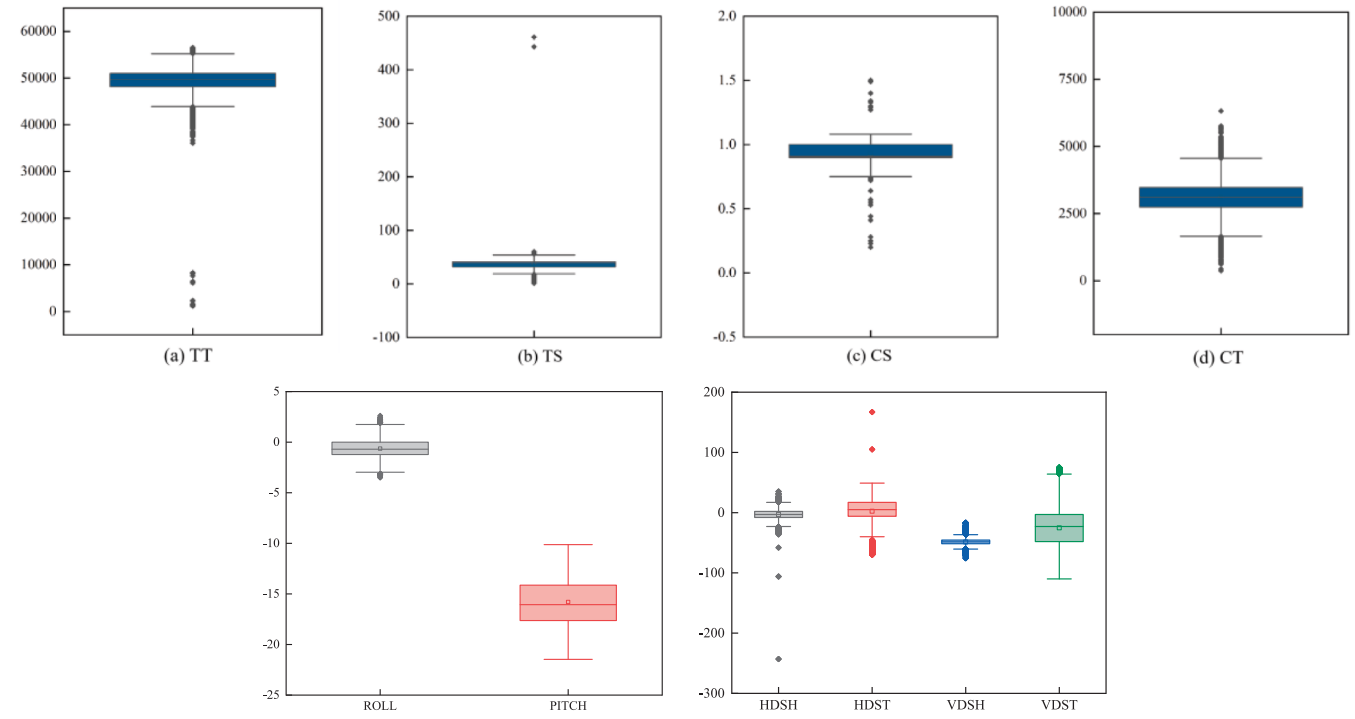


Fig. 5. Outlier detection using boxplot analysis.

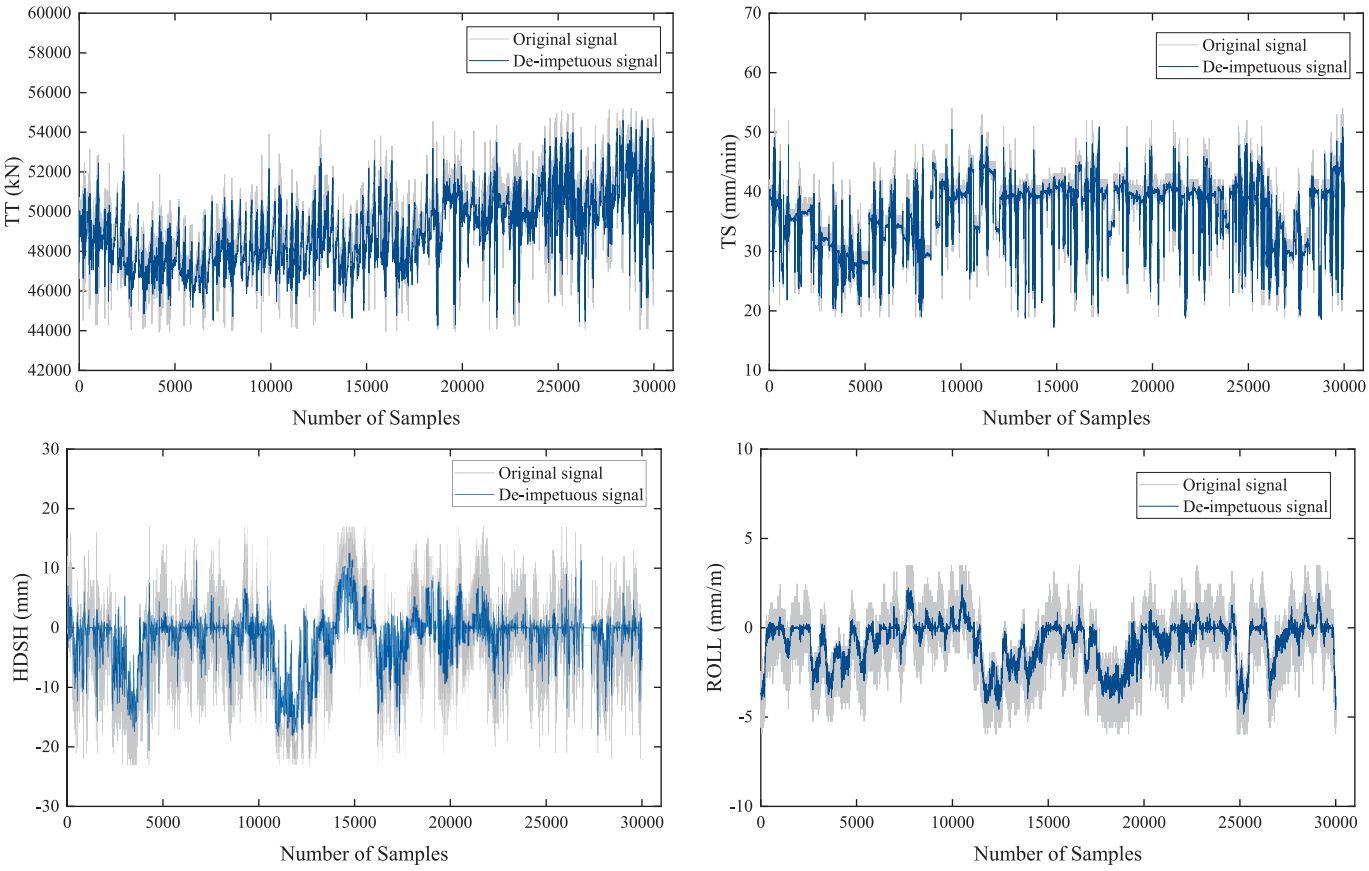


Fig. 6. Signal characteristics before and after denoising.

parameters for optimization. These include the pressure of thrust cylinder pressure groups A–F, total thrust, thrust speed, cutterhead speed, cutterhead torque, air cushion pressure, grouting pressure, inlet slurry density, outlet slurry density, inlet slurry flow rate, and outlet slurry



Fig. 7. (a) Geometric model developed by SolidWorks; (b) User interface.

flow rate.

After selecting the 16 most relevant parameters, further analysis was performed to investigate how varying parameter subsets affect optimization performance. Specifically, the study explored whether a "fewer parameters, better performance" strategy could be viable. Based on their average SHAP values, the parameters were grouped into three scenarios: Scenario 1 includes only the top 50%, representing the most influential parameters; Scenario 2 adds parameters up to the top 80%, covering medium- to high-importance factors; and Scenario 3 includes all parameters. The adjustable parameters included in each scenario are presented in Fig. 9. This stepwise expansion strategy provides a basis for evaluating the trade-offs between model complexity and control accuracy.

3.4.2. *Hyparameter setting of ABC-NSGA-III*

To balance convergence stability with global search capability, the key hyperparameters of NSGA-III were determined through the grid search procedure described in Section 2.3.2. The full set of hyperparameters is summarized in Table 3.

To ensure the feasibility and practicality of optimized solutions, a rule-based model was established within the DT virtual layer. This model defines the allowable value ranges for the 16 selected parameters based on historical construction data, the *Code for Construction and Acceptance of Shield Tunneling* (GB 50446–2017), and insights from relevant studies (Huang et al., 2022; Zhou et al., 2019). The constraint rules serve to narrow the feasible domain of tunneling parameters, thereby aligning algorithmic outputs with actual construction requirements. The detailed constraint ranges are presented in Table 4.

3.4.3. *Comparative analysis of optimization results*

To evaluate the performance of ABC-NSGA-III across three distinct operational scenarios, an optimization rate index was used to quantify improvements in shield attitude-related objectives before and after optimization. Each evaluation is based on the average results of 10 independent runs. As shown in Table 5, ABC-NSGA-III delivers strong improvements across most deviation parameters, particularly in controlling VDSH, where the optimization rate reaches over 28.01%. Moreover, the optimal attitudes obtained under each scenario illustrate the correlation between the dimensionality of parameter optimization and overall performance. As the number of optimized parameters increases, the overall optimization rate rises substantially, reaching 32.02% in Scenario 3. This indicates that a higher-dimensional parameter space allows the algorithm to better exploit solution possibilities and improve shield attitude control. Comparison of optimal solutions in the Pareto sets shows that Scenario 3 outperforms Scenarios 1 and 2 across all objectives. This confirms that regulating all 16 tunneling parameters simultaneously achieves the best attitude control. It also validates the reliability of the parameter importance ranking from SHAP analysis and highlights the feasibility of ABC-NSGA-III for shield attitude control.

Based on the results of Scenario 3, a high-performance set of Pareto-optimal solutions was obtained (see Table 6). These solutions represent various optimal combinations of shield tunneling parameters that balance the multiple attitude-related objectives. Through analysis of solution distributions, recommended control ranges for each parameter were established. The optimal parameter combinations derived from the MOO was then fed back through the data interaction layer to initialize the control rules. The DT-based framework facilitated closed-loop feedback, dynamically adjusting parameters based on the evolving relationship between machine behavior and control settings. This enables continuous refinement of key parameters throughout the entire tunneling process.

4. Discussions

4.1. *Comparative performance analysis*

The performance of ABC-NSGA-III was further benchmarked against the standard NSGA-III and PSO-NSGA-III. The evaluation was conducted using six identical optimization models, which tasked with minimizing six attitude deviations. The results, summarized in Table 7, demonstrate the superiority of ABC-NSGA-III. Compared with the standard NSGA-III, ABC-NSGA-III exhibited enhanced global exploration and local exploitation capabilities. The average optimization magnitudes of HDSH and VDSH improved by 6.12% and 17.13%, respectively, and the best-found optimal values for HDSH and VDSH were enhanced by 4.74% and 14.98%. Compared with PSO-NSGA-III, ABC-NSGA-III yielded a 2.5% and 8.14% increase in the average performance for HDSH and VDSH, and a 2.21% and 7.34% improvement in their respective optimal values.

To compare the convergence performance of the three algorithms under the 100-ring operational dataset, a sliding-window approach was employed. Convergence was defined as the condition in which fluctuations of the HV value over the most recent 20 generations fell below 1%, and the slope of the fitted linear regression approximated zero. The results are reported in Table 8. From the perspective of near real-time performance, ABC-NSGA-III demonstrates a distinct advantage in convergence speed. The algorithm converges in approximately 900.5 s, yielding a stable Pareto-optimal solution set, with the convergence phase accounting for only 64% of the total computational time. By contrast, the standard NSGA-III requires 993.6 s, with convergence occupying 92% of its iteration time. Moreover, ABC-NSGA-III achieves an HV value of 0.532, representing a 35% improvement over that of the standard NSGA-III (0.394), thereby confirming its capability to produce high-quality solutions within constrained timeframes. The intermediate performance of PSO-NSGA-III, with a convergence time of 945.0 s and an HV value of 0.482, highlights the effectiveness of the ABC search

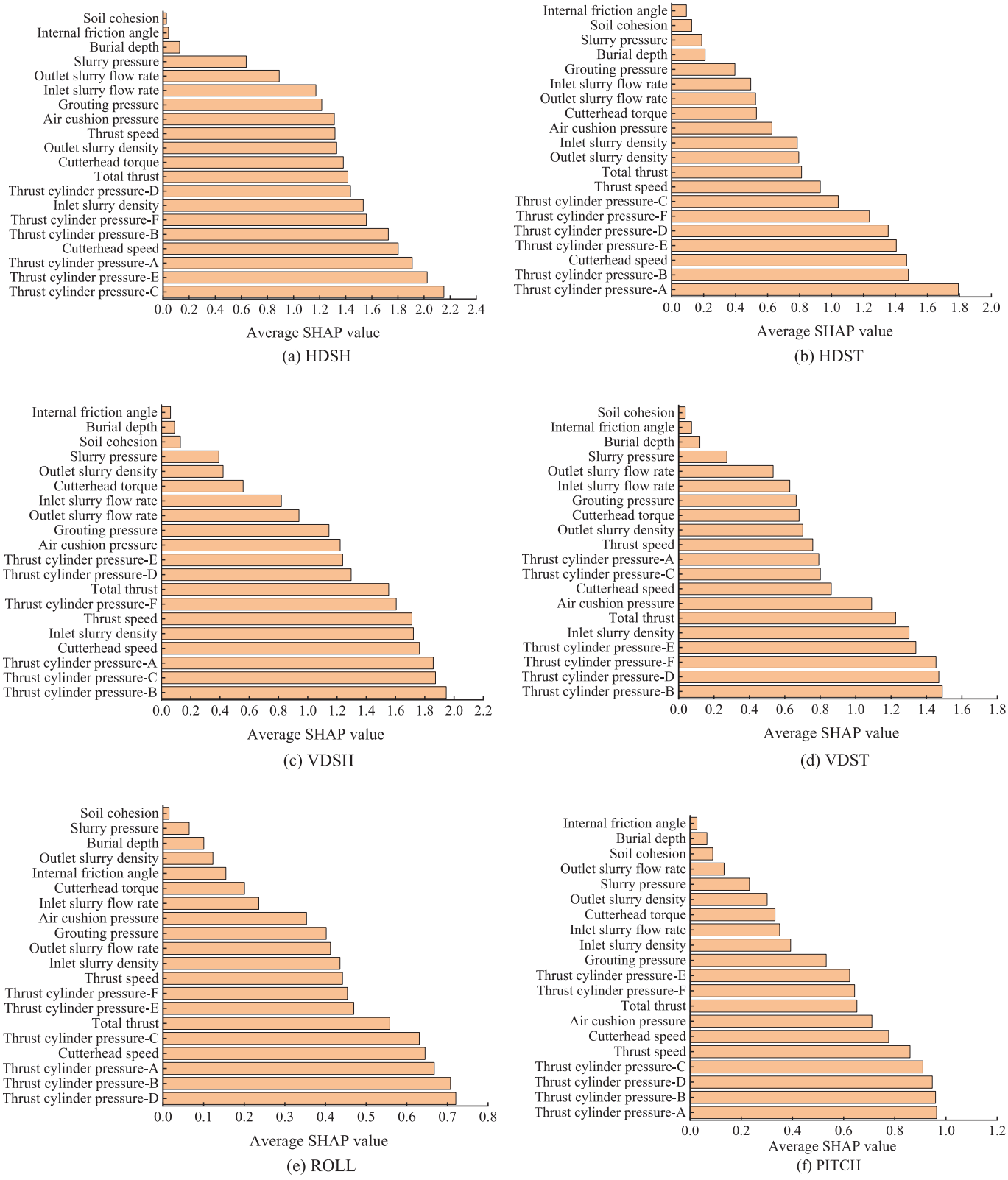


Fig. 8. SHAP analysis of tunneling parameters on shield attitude.

strategy in achieving a balance between convergence performance and solution quality.

Regarding computational efficiency, the ABC–NSGA-III module requires approximately 900 s on average to complete optimization and model updating, constituting the dominant share of the framework's total response time. In contrast, delays stemming from data acquisition and communication—arising from 10-second sampling intervals and roughly 500-millisecond transmission latency—can be negligible in the overall workflow. In the case study project, according to the "*Shanghai Municipal Engineering Budget Quota Volume 1: Road, Bridge, and Tunnel*

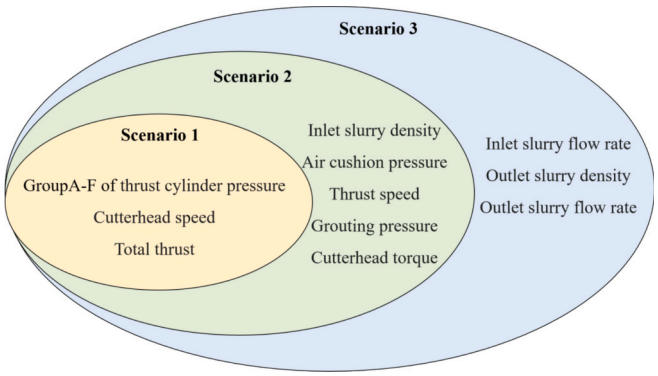


Fig. 9. Parameters adjusted in each scenario.

Table 3
Configurations of ABC-NSGA-III.

Algorithm module	Parameter	Value
NSGA-III	Number of objective functions	6
	Population size	200
	Crossover probability	0.7
	Mutation probability	0.01
	Maximum number of iterations for NSGA-III	300
	Number of bees	200
ABC	Number of iterations for ABC	50
	Nectar collection limit (abandonment threshold)	100

Table 4
Range of tunneling parameters.

Parameter	Range	Unit
Thrust cylinder pressure-A	[12.48, 70.14]	bar
Thrust cylinder pressure-B	[10.91, 68.05]	bar
Thrust cylinder pressure-C	[75.04, 115.56]	bar
Thrust cylinder pressure-D	[77.33, 122.47]	bar
Thrust cylinder pressure-E	[77.51, 118.66]	bar
Thrust cylinder pressure-F	[9.85, 74.39]	bar
Total thrust	[44061.14, 54661.67]	kN
Thrust speed	[17.12, 51.84]	mm/min
Cutterhead speed	[0.75, 1.08]	r/min
Cutterhead torque	[1482.49, 4252.730]	kN ·m
Inlet slurry density	[1.30, 1.34]	g/cm^3
Outlet slurry density	[1.23, 1.38]	g/cm^3
Inlet slurry flow rate	[1349, 1870]	m^3/h
Outlet slurry flow rate	[1514, 2169]	m^3/h
Air cushion pressure	[2.97, 3.12]	bar
Grouting pressure	[2.37, 7.12]	bar

Engineering (SHA1-31(01)-2016)", the construction cycle for a single ring is 90–120 min (5400–7200 s). The time consumed by the optimization process thus accounts for only 12.5%-16.7% of the total cycle. This proportion confirms that the proposed framework can complete all required computations within the available construction interval prior to excavation of the subsequent tunnel segment, thereby supporting dynamic control of tunnel construction.

4.2. Prospection of DT in tunnel construction

Using the Shanghai Airport Link Line project as a case study, this study illustrates the efficacy of the proposed framework for optimizing shield tunneling parameters. Unlike conventional approaches that rely on operator experience (Liu et al., 2021) and current DT applications that remain largely limited to monitoring (Zheng et al., 2024), the proposed framework represents a step forward by integrating DT technology with advanced optimization algorithms.

The framework algorithmically generates optimal parameter combinations in an interpretable form, providing a quantitative basis to support operator decisions. Within this human–machine collaborative mechanism, the framework performs MOO, while operators—drawing on their judgment of complex conditions and expert knowledge—evaluate the optimization results. This mechanism treats operator expertise as a safeguard, filtering out abnormal outputs that may arise from field uncertainties. In doing so, the framework fuses the computational efficiency of data-driven models with the adaptive intelligence of human expertise, which ensures its applicability under real-world tunneling conditions.

Table 5
Comparison of optimal values in different scenarios.

Optimization objective		Scenario			
		Average monitoring value	Scenario 1	Scenario 2	Scenario 3
HDSH (mm)	Optimal value (Optimization rate)	2.79	2.63 (5.73%)	2.36 (15.41%)	2.21 (20.79%)
HDST (mm)	Optimal value (Optimization rate)	12.72	11.86 (13.56%)	10.35 (24.56%)	9.47 (30.98)
VDSH (mm)	Optimal value (Optimization rate)	46.77	33.67 (28.01%)	30.44 (34.92%)	27.76 (40.65%)
VDST (mm)	Optimal value (Optimization rate)	29.36	26.39 (10.12%)	23.65 (19.45%)	22.41 (23.67%)
ROLL (mm/m)	Optimal value (Optimization rate)	0.53	0.52 (1.89%)	0.44 (16.98%)	0.39 (26.42)
PITCH (mm/m)	Optimal value (Optimization rate)	15.81	14.33 (9.36%)	14.06 (11.07%)	13.24 (16.26%)
Overall optimization rate		–	20.97%	28.31%	32.02%

The hierarchical structure of the proposed framework provides a for scalability and broader applicability. Future work may extend this framework to more diverse geological conditions. In hard-rock strata, for instance, the optimization objectives could be reformulated to balance cutter wear and advance rate through integration with appropriate geomechanical models; When passing through aquifer-bearing layers, the framework could instead prioritize tunnel-face stability and surface-settlement control. These extensions underscore the framework's potential to incorporate context-specific objectives to meet the needs of different tunneling projects.

5. Conclusion

This paper proposes a framework for optimizing shield tunneling parameters by integrating DT technology with an improved NSGA-III. The framework facilitates the transmission of data from the physical layer to the data interaction layer, thereby supporting digital modeling and optimization decision-making. Within the application service layer, key tunneling parameters are identified using SHAP method. These parameters are then grouped into multiple optimization scenarios based on their influence on tunneling performance. A MOO process is carried out using ABC-NSGA-III to improve global search diversity and solution quality. The optimized parameter sets are subsequently transmitted back through the data interaction layer to guide physical shield tunneling operations.

The feasibility and effectiveness of the proposed framework were validated using real-world project data. Six shield attitude indicators were optimized across three defined parameter scenarios. Through SHAP analysis, 16 critical tunneling parameters were identified. In the full-parameter optimization scenario, the framework achieved an overall optimization rate of 32.02%. Compared with the standard NSGA-III and PSO-NSGA-III algorithms, ABC-NSGA-III demonstrated faster

Table 6
Pareto optimal tunneling parameters and their control ranges in Scenario 3.

Parameter name	Max. value	Min. value	Ave. value	Optimal solution	Recommended control range
Thrust cylinder pressure-A (bar)	42.57	34.11	38.59	37.54	34.11 ~ 42.57
Thrust cylinder pressure-B (bar)	63.67	55.79	59.44	60.49	55.79 ~ 63.67
Thrust cylinder pressure-C (bar)	114.75	77.98	107.94	103.22	77.98 ~ 114.75
Thrust cylinder pressure-D (bar)	117.67	90.13	101.50	107.07	90.13 ~ 117.67
Thrust cylinder pressure-E (bar)	115.09	79.92	95.49	88.42	79.92 ~ 115.09
Thrust cylinder pressure-F (bar)	63.17	26.01	41.23	54.03	26.01 ~ 63.17
Total thrust (KN)	51607.89	46545.95	49154.79	46670.20	46545.95 ~ 51607.89
Thrust speed (mm/min)	35.87	19.05	24.45	29.88	19.05 ~ 35.87
Cutterhead speed (r/min)	1.08	0.79	0.89	0.84	0.79 ~ 1.08
Cutterhead torque (KN ·m)	3021.98	2225.52	2671.54	2583.61	2225.52 ~ 3021.98
Inlet slurry density (g/cm^3)	1.34	1.30	1.32	1.34	1.30 ~ 1.34
Outlet slurry density (g/cm^3)	1.38	1.24	1.28	1.31	1.24 ~ 1.38
Inlet slurry flow rate (m^3/h)	1693.83	1651.35	1675.14	1680.39	1651.35 ~ 1693.83
Outlet slurry flow rate (m^3/h)	2144.39	1742.44	1915.28	1935.48	1742.44 ~ 2144.39
Air cushion pressure (bar)	3.10	2.96	3.02	2.99	2.96 ~ 3.10
Grouting pressure (bar)	6.68	2.77	4.78	4.86	2.77 ~ 6.68

convergence and higher solution quality, reducing shield attitude deviation by 2.21%–17.13%. These results confirm the practical viability and performance advantages of the proposed framework in tunnel construction.

Future studies can further enhance the practical applicability of the proposed framework in complex tunnel construction. First, while this study employed the DWT to mitigate measurement noise, ongoing advances in sensing technologies offer opportunities to improve data fidelity at the source. Second, techniques such as transfer learning hold promise for strengthening the model's generalization capability across different geological conditions. Finally, training programs for operators should be established to deepen their understanding of the proposed framework and facilitate its effective implementation in field operations.

Table 7
Comparison between the average and optimal values of the objective function.

Evaluation indexes		ABC-NSGA-III	NSGA-III (Improvement degree)	PSO-NSGA-III (Improvement degree)
HDSH (mm)	Ave.	2.30	2.45 (6.12%)	2.36 (2.50%)
	Opt.	2.21	2.32 (4.74%)	2.26 (2.21%)
HDST (mm)	Ave.	9.58	11.36 (15.67%)	10.28 (6.81%)
	Opt.	9.47	10.95 (13.52%)	9.98 (5.11%)
VDSH (mm)	Ave.	28.34	34.20 (17.13%)	30.85 (8.14%)
	Opt.	27.76	32.65 (14.98%)	29.96 (7.34%)
VDST (mm)	Ave.	22.62	26.48 (14.58%)	24.25 (6.72%)
	Opt.	22.41	25.30 (11.42%)	23.68 (5.36%)
ROLL (mm/m)	Ave.	0.41	0.46 (10.87%)	0.44 (6.82%)
	Opt.	0.39	0.42 (7.14%)	0.41 (4.88%)
PITCH (mm/m)	Ave.	13.63	14.93 (8.71%)	14.18 (3.88%)
	Opt.	13.24	14.67 (9.75%)	13.75 (3.71%)

Table 8
Comparison of computational efficiency and optimization effectiveness.

Algorithm	Computation time (s)	HV value
NSGA-III	993.6	0.394
PSO-NSGA-III	945.0	0.482
ABC-NSGA-III	900.5	0.532

CRediT authorship contribution statement

Congzhen Yang: Writing – original draft, Software, Methodology. **Zhikun Ding:** Writing – review & editing, Supervision. **Tianrui Liu:** Visualization, Software. **Zezhou Wu:** Writing – review & editing, Visualization. **Ke Chen:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This study is supported by National Natural Science Foundation of China (U21A20151), and Open Research Fund of Key Laboratory of Coastal Urban Resilient Infrastructures (Shenzhen University), Ministry of Education (CURI-G202402).

Data availability

Data will be made available on request.

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